

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR10074 COMPUTER COMMUNICATIONS AND NETWORKS

May 2020

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

Answer any TWO of the three questions. If more than two questions are answered, only QUESTION 1 and QUESTION 2 will be marked.

All questions carry equal weight.

This is an OPEN BOOK examination.

Year 3 Courses

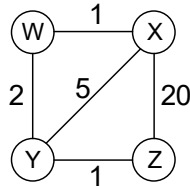
Convener: S.Ramamoorthy

External Examiners: S.Rogers, S.Kalvala, H.Vandierendonck

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Under what circumstances does non-persistent HTTP outperform persistent HTTP in terms of web page download time? [2 marks]
- (b) Consider a home Internet access scenario. To speed up web page download times and make efficient use of your Internet access connection, you decide to enable caching within your web browser. How does this compare with the alternative of web caching proxy server installed within the infrastructure of your Internet Service Provider (ISP)? [3 marks]
- (c) Consider the following notation to model client-side buffering for video streaming. Let B denote the size of the client's application buffer in bits. Let $Q(< B)$ denote the number of bits that must be buffered before the client application begins playout. Let r denote the video consumption rate during playback. Finally, assume that the client receives data into the application buffer from the network at a rate x (referred to as the *fill rate*) when the client buffer is not full.
 - i. When $x < r$, determine the length of each continuous playout and freezing period as a function of Q , r and x . [6 marks]
 - ii. If $x > r$, then at what time, say t_f , does the client application buffer become full? [3 marks]
- (d) With the increasing popularity of video content, Internet Service Providers (ISPs) face the challenge of managing traffic spikes during peak hours. If you were an ISP, what policy would you apply to shape the traffic of high bandwidth consuming users at peak hours and explain how you would implement it? Suppose that some of those users are willing to pay an extra premium to avoid such traffic shaping. How would you support your premium users? [5 marks]
- (e) Outline three key mechanisms used for providing soft quality of service (QoS) guarantees via differentiated service. [6 marks]

2. (a) Distinguish between control and data plane at the network layer. Mention two different paradigms to realise the network layer control plane and the same for the data plane. [4 marks]
- (b) List five key differences between IPv4 and IPv6. [5 marks]
- (c) Consider the following network consisting of 4 nodes. An integer on an edge denotes communication cost between two nodes.



Using the distance vector algorithm, complete the following routing tables. To get you started, initial routing tables of round 0 are shown below. [12 marks]

Round 0:

		W table				X table				Y table				Z table			
		cost to				cost to				cost to				cost to			
		w	x	y	z	w	x	y	z	w	x	y	z	w	x	y	z
from	w	0	1	2	∞	w	∞	∞	∞	w	∞	∞	∞	w	∞	∞	∞
	x	∞	∞	∞	∞	x	1	0	5	x	∞	∞	∞	x	∞	∞	∞
	y	∞	∞	∞	∞	y	∞	∞	∞	y	2	5	0	y	∞	∞	∞
	z	∞	∞	∞	∞	z	∞	∞	∞	z	∞	∞	∞	z	∞	20	1

Round 1:

		cost to				cost to				cost to				cost to			
		w	x	y	z	w	x	y	z	w	x	y	z	w	x	y	z
from	w					w				w				w			
	x					x				x				x			
	y					y				y				y			
	z					z				z				z			

Round 2:

		cost to				cost to				cost to				cost to			
		w	x	y	z	w	x	y	z	w	x	y	z	w	x	y	z
from	w					w				w				w			
	x					x				x				x			
	y					y				y				y			
	z					z				z				z			

Round 3:

		cost to				cost to				cost to				cost to			
		w	x	y	z	w	x	y	z	w	x	y	z	w	x	y	z
from	w					w				w				w			
	x					x				x				x			
	y					y				y				y			
	z					z				z				z			

- (d) Route oscillations occur in (vanilla) link-state and distance-vector routing algorithms when the link cost metric is congestion dependent. Describe an approach to avoid such oscillations while still using congestion based link cost metric.

[4 marks]

3. (a) Discuss two differences between a router and an Ethernet switch. [4 marks]
- (b) Outline self-learning in a switch. [1 mark]
- (c) How does self-learning help in the case of a host moving between two access points, both connected to a common switch via two different ports at the switch? [2 marks]
- (d) Consider the 4-bit generator, $G = 1001$, and suppose that the data portion of a bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 10011101. Show the actual bit stream transmitted. Suppose that the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end. Give an example of bit errors in the bit stream transmitted that will not be detected by the receiver. [6 marks]
- (e) Consider the effect of using slow-start on a link with a 10ms round-trip time and no congestion. The receive window is 24KB and the maximum segment size is 2KB. How long does it take before the first full window can be sent in one transmission round? [2 marks]
- (f) Consider sending a large file from one host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start. Assume *CongWin* increases by 1 MSS (maximum segment size) every time a batch of ACKs is received. Also assume approximately constant RTTs (round-trip times).
 - i. How long does it take for *CongWin* to increase from 1 MSS to 7 MSS (assuming no loss events)? [3 marks]
 - ii. What is the average throughput (in terms of MSS and RTT) for this connection over that period (from sub-part (i))? [1 mark]
- (g) Suppose a host has a 1-MB file that is to be sent to another host over a 2-Mbps link with 10 ms one-way propagation delay. The file takes 1 second of CPU time to compress 30%, or 2 seconds to compress 50%. Ignore queueing delays.
 - i. Calculate the bandwidth at which the combined time taken for compression and transmission for each of the compression options equals that of the uncompressed case. [4 marks]
 - ii. Does propagation delay also get affected? Why or why not? [2 marks]